

Fundamentals of Programming & Computer Science CS 15-112

Lists I

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1D List & Aliasing

Practice- Figure out the Bug

```
1 #There's a bug in the following code!
2
3 def destructiveRemoveNonIntsWrong(L):
4     for e in L:
5         print("Checking", e)
6         if not isinstance(e, int):
7             L.remove(e)
8
9 L = [112, '15', 4, 2, 'hey', 15.112]
10 assert(destructiveRemoveNonIntsWrong(L) == None)
11 print(L) # it never checks 4 or 15.112. why?
12 assert(L == [112, 4, 2])
```

FR Practice

(a) Write the function `nondestructiveRemoveEvens (L)` that returns a **new list** containing only the odd values from `L`. The original list `L` must remain unchanged.

(b) Write the function `destructiveRemoveEvens (L)` that **removes all even values from `L` in place**. This function should modify `L` directly and return `None`.

You may assume `L` is a list of integers.